

REG NO: 22TD0006 NAME: ARUL SHRINIVAS A YEAR/SEM/SEC: Y3/S5/B

PREDEFINED EXCEPTION

PROGRAM:

```
SQL>set serveroutput on;
      declare
      empid number(8);
      emname varchar(10);
      begin
      empid:=&empid;
      select empname into empname from employee where empno=empid;
      dbms_output.put_line('The employee name is:'||empname);
      exception
      when no_data_found then
      dbms_output.put_line('data not found');
      end;
      /
```

OUTPUT:

```
Enter value for empid: 1
old 5: empid:=&empid;
new 5: empid:=1;
the employe name is:radha

PL/SQL procedure successfully completed.

SQL> /
Enter value for empid: 5
old 5: empid:=&empid;
new 5: empid:=5;
data not found

PL/SQL procedure successfully completed.
```

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USER DEFINED EXCEPTION

PROGRAM:

```
SQL> set serveroutput on;

declare
co number;
exceeds_value exception;
begin
select count(*)into co from bankgrd;
if co<5 then
insert into BANK values(&s_no,'&cust_name',&balance);
else
raise exceeds_value;
end if;
exception
when exceeds_value then
dbms_output.put_line('row will not be inserted');
end;
/
```

OUTPUT:

```
Enter value for s_no: 5
Enter value for cust_name: kaviya
Enter value for balance: 80000
old 7: insert into BANK values(&s_no,&cust_name,&balance);
new 7: insert into BANK values(5,'kaviya',80000);

PL/SQL procedure successfully completed.
```

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DATA DEFINITION LANGUAGE

CREATE TABLE COMMAND

Syntax:

SQL> Create table<table name>(fieldname1 datatype1(size),fieldname2 datatype2(size)....fieldname n datatype n(size));

Query:

```
SQL> create table empxd(empno number(5), emp_name char(20), salary varchar(7), age number(4));  
  
Table created.
```

Created table:

```
SQL> desc empxd;  
Name                               Null?      Type  
-----  
EMPNO                               NUMBER(5)  
EMP_NAME                            CHAR(20)  
SALARY                             VARCHAR2(7)  
AGE                                NUMBER(4)
```

ALTER COMMAND

The Alter command is associated with the following commands

- ADD
- MODIFY
- DROP

USES OF ALTER COMMANDS

1. We can add a new field in an already created table
2. We can change the data type and size in a already created table
3. We can add integrity constraints to a field
4. We can drop integrity constraints of a field
5. We can drop a field (or) column from a table

ADD FIELD

Syntax:

*SQL>alter table<table_name>add (fieldname1
datatype1(size)....fieldname n
datatype n(size));*

Query:

```
SQL> alter table empxd add(job varchar(20));
```

```
Table altered.
```

```
SQL> desc empxd;
```

Name	Null?	Type
EMPNO		NUMBER(5)
EMP_NAME		CHAR(20)
SALARY		VARCHAR2(7)
AGE		NUMBER(4)
JOB		VARCHAR2(20)

MODIFY FIELD

Syntax:

SQL> alter table <table_name> modify (fieldname1 datatype1(size).....fieldname n datatype n(size));

Query:

```
SQL> alter table empzd modify(emp_name char(30));  
Table altered.  
SQL> desc empzd;  
Name                               Null?    Type  
-----  
EMPNO                               NUMBER(5)  
EMP_NAME                            CHAR(30)  
SALARY                              VARCHAR2(7)  
AGE                                 NUMBER(4)  
JOB                                 VARCHAR2(20)
```

DROP FIELD

Syntax:

SQL> alter table <table_name> drop <column_name>;

Query:

```
SQL> alter table empzd modify(emp_name char(30));  
Table altered.  
SQL> desc empzd;  
Name                               Null?    Type  
-----  
EMPNO                               NUMBER(5)  
EMP_NAME                            CHAR(30)  
SALARY                              VARCHAR2(7)  
AGE                                 NUMBER(4)  
JOB                                 VARCHAR2(20)
```

TRUNCATE TABLE

Syntax:

SQL> truncate table<table_name>;

Query:

```
SQL> truncate table empxd;  
  
Table truncated.  
  
SQL> select * from empxd;  
  
no rows selected
```

RENAME TABLE

Syntax:

SQL> rename<table_name> to <new_table_name>;

Query:

```
SQL> rename empxd to employeexd;  
  
Table renamed.  
  
SQL> desc employeexd;  
Name                               Null?      Type  
-----  
EMPNO                               NUMBER(5)  
EMP_NAME                           CHAR(30)  
SALARY                             VARCHAR2(7)  
AGE                                 NUMBER(4)
```


DROP TABLE COMMAND

Syntax:

SQL> drop table<table_name>;

Query:

 Select Run SQL Command Line

```
SQL> drop table employee;  
Table dropped.
```

After drop the table will not be present,

 Select Run SQL Command Line

```
SQL> select * from employee;  
select * from employee  
      *  
ERROR at line 1:  
ORA-00942: table or view does not exist
```

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DATA MANIPULATION LANGUAGE

INSERT

Syntax:

SQL> insert into <table_name> values<value1,value2....value n>;

Query:

```
SQL> insert into e1(e_id,e_name,e_sal)values(101,'raj',1000);
1 row created.

SQL> insert into e1(e_id,e_name,e_sal)values(102,'jessi',2000);
1 row created.

SQL> insert into e1(e_id,e_name,e_sal)values(103,'john',3000);
1 row created.
```

```
SQL> select * from e1;

      E_ID E_NAME      E_SAL
-----
      101 raj         1000
      102 jessi        2000
      103 john         3000
```

UPDATE

Syntax:

*SQL> update<table_name> set field_name=<expression>where
[condition];*

Query:

```
SQL> update e1 set e_sal=5000 where e_name='john';  
1 row updated.
```

DELETE

Syntax:

SQL> delete from <table_name>where [condition];

Query:

```
SQL> delete from e1 where(e_id=103);  
1 row deleted.
```

```
SQL> select * from e1;  
  
      E_ID E_NAME      E_SAL  
-----  
      101 raj         1000  
      102 jessi        2000
```

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DATA QUERY LANGUAGE

CREATE TABLE employe1:

```
SQL> create table employe1(empno number(5),ename char(10),job char(10),MGR number(5),sal number(5),deptno number(5),address char(10));

Table created.

SQL> insert into employe1(empno,ename,job,MGR,sal,deptno,address)values(7369,'guna','manager',7902,2000,20,'chennai');

1 row created.

SQL> insert into employe1(empno,ename,job,MGR,sal,deptno,address)values(7499,'kaushi','salesman',7617,1600,30,' ');

1 row created.
```

TABLE STRUCTURE:

```
SQL> select*from employe1;
```

EMPNO	ENAME	JOB	MGR
7369	guna	manager	7902
2000	20 chennai		
7499	kaushi	salesman	7617
1600	30		
7521	ash	clerk	7839
2975	10 trichy		

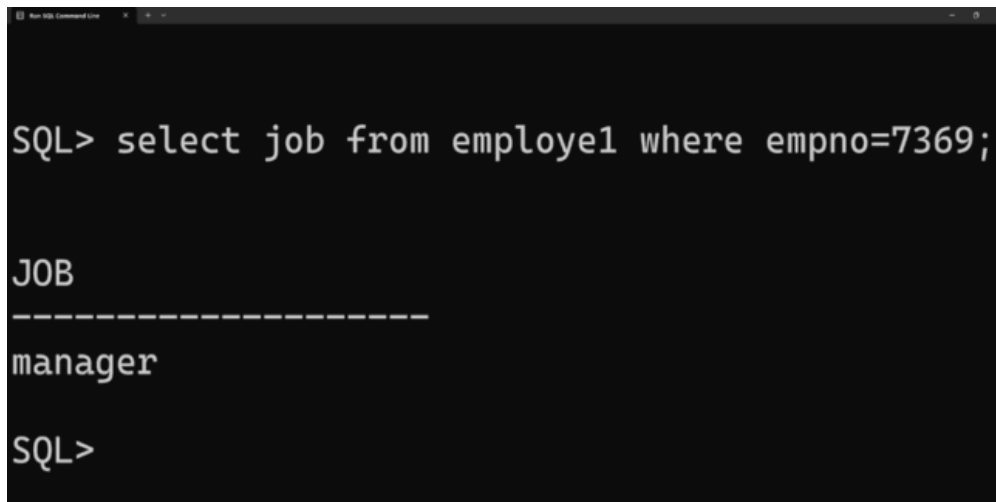
EMPNO	ENAME	JOB	MGR
7342	suman	salesman	7893
1200	20 hyderabad		
76563	saran	manager	7964
4000	30		

SELECT

Syntax:

SQL>select <column_name>from<table_name>where[condition];

Query:



```
SQL> select job from employee1 where empno=7369;

JOB
-----
manager

SQL>
```

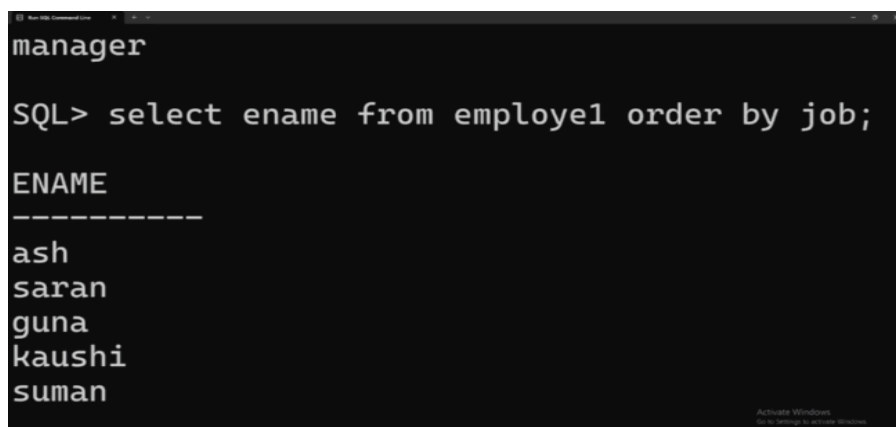
ORDER BY

➤ ASCENDING:

Syntax:

*SQL>select<column_name>from<table_name>order
by<column_name>;*

Query:



```
manager

SQL> select ename from employee1 order by job;

ENAME
-----
ash
saran
guna
kaushi
suman
```

➤ **DESCENDING:**

Syntax:

*SQL>select<column_name>from<table_name>order
by<column_name>desc;*

Query:

```
SQL> select ename from employee1 order by job desc;

ENAME
-----
kaushi
suman
guna
saran
ash

SQL> |
```

GROUP BY

Syntax:

*SQL>select<column_name>from<table_name>order
by<column_name>;*

Query:

```
SQL> select min(salary)from employee1 group by deptno;

MIN(SALARY)
-----
          1600
          1200
          2975
```

HAVING CLAUSE

Syntax:

SQL>select<column_name>from<table_name>where[condition]group by column having[condition];

Query:

```
SQL> select min(deptno)from employe1 group by salary having count(deptno)>=1;

MIN(DEPTNO)
-----
          30
          20
          20
          10
          30

SQL>
```

NULL

Syntax:

SQL>select<column_name>from<table_name>where<column_name>is null;

SQL>select<column_name>from<table_name>where<column_name>is not null;

Query:

```
SQL> select ename from employe1 where address=' ';

ENAME
-----
kaushi
saran
```

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DATA CONTROL LANGUAGE

CREATE TABLE employee1 AND employee2:

```
SQL> create table employee1(SLNO number(6),EMP_ID varchar(10),EMPNAME char(10),DESIGNATION char(12));
Table created.

SQL> insert into employee1(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(1,'19TD0301','jack','analyst');
1 row created.

SQL> insert into employee1(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(2,'19TD0302','harry','manager');
1 row created.

SQL> create table employee2(SLNO number(6),EMP_ID varchar(10),EMPNAME char(10),DESIGNATION char(12));
Table created.

SQL> insert into employee2(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(2,'19TD0302','harry','manager');
1 row created.

SQL> insert into employee2(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(5,'19TD0305','morris','fresher');
1 row created.
```

TABLE STRUCTURE:

```
mysql> select * from employee1;
+-----+-----+-----+-----+
| SLNO | EMPID | empname | designation |
+-----+-----+-----+-----+
| 1 | 22TD0055 | jack | analyst |
| 2 | 22TD0301 | harry | manager |
| 3 | 22TD0211 | john | tester |
| 4 | 22TD0050 | peter | developer |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

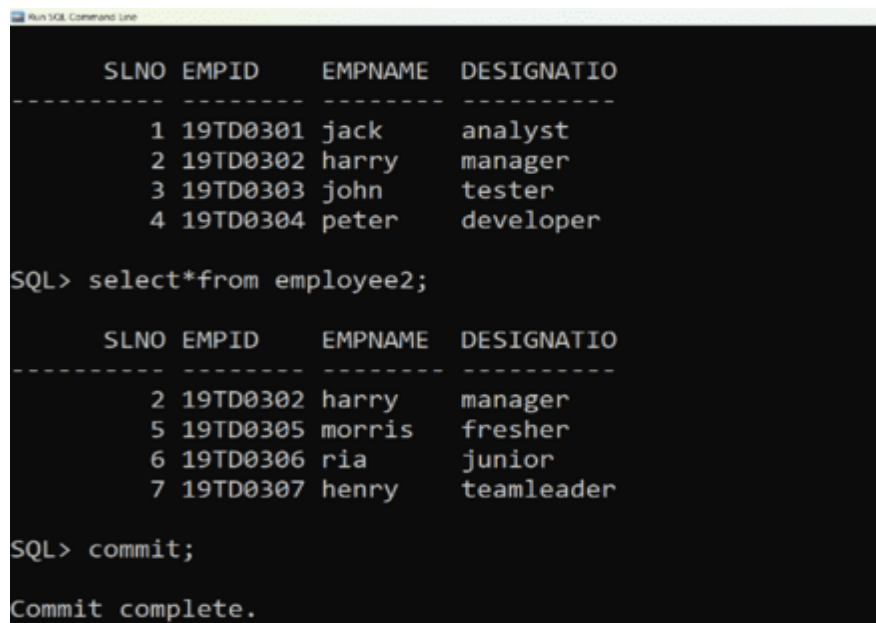
mysql> select * from employee2;
+-----+-----+-----+-----+
| SLNO | EMPID | empname | designation |
+-----+-----+-----+-----+
| 2 | 22TD0302 | harry | manager |
| 5 | 22TD0305 | morris | fresher |
| 6 | 22TD0306 | ria | junior |
| 7 | 22TD0307 | henry | teamleader |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```


COMMIT

Syntax:

SQL> commit;

Query:



```
Run SQL Command Line

      SLNO  EMPID      EMPNAME  DESIGNATIO
-----
      1 19TD0301  jack      analyst
      2 19TD0302  harry      manager
      3 19TD0303  john       tester
      4 19TD0304  peter      developer

SQL> select*from employee2;

      SLNO  EMPID      EMPNAME  DESIGNATIO
-----
      2 19TD0302  harry      manager
      5 19TD0305  morris     fresher
      6 19TD0306  ria        junior
      7 19TD0307  henry      teamleader

SQL> commit;

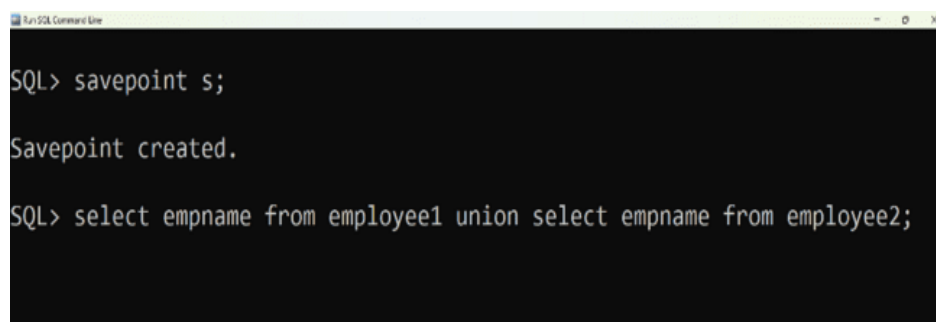
Commit complete.
```

SAVEPOINT

Syntax:

SQL> savepoint <name>;

Query:



```
Run SQL Command Line

SQL> savepoint s;

Savepoint created.

SQL> select empname from employee1 union select empname from employee2;
```

ROLLBACK

Syntax:

SQL> rollback;

Query:

```
Run SQL Command Line

SQL> roll back;
Rollback complete.
SQL> select*from employee1;

      SLNO  EMPID      EMPNAME  DESIGNATIO
-----
      1 19TD0301  jack      analyst
      2 19TD0302  harry      manager
      3 19TD0303  john      tester
      4 19TD0304  peter     developer

SQL> select*from employee2;

      SLNO  EMPID      EMPNAME  DESIGNATIO
-----
      2 19TD0302  harry      manager
      5 19TD0305  morris     fresher
      6 19TD0306  ria        junior
      7 19TD0307  henry     teamleader
```

GRANT

Syntax:

SQL>grant<select> on <table_name>to user;

Query:

```
Run SQL Command Line

SQL> grant all on employee1 to system;

Grant succeeded.

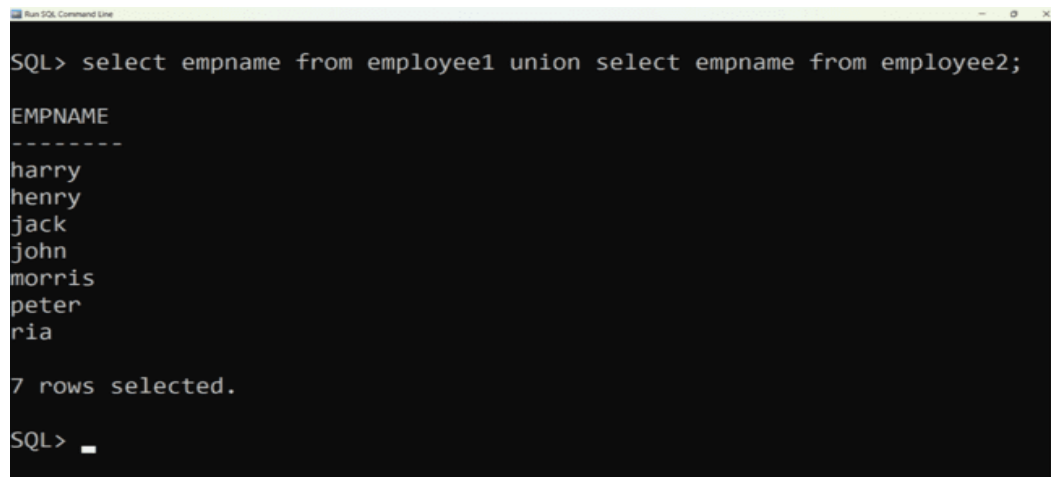
SQL> _
```

UNION

Syntax:

*SQL>select<column_name>from<table_name>union select
<column_name>from<table_name>;*

Query:



```
SQL> select empname from employee1 union select empname from employee2;

EMPNAME
-----
harry
henry
jack
john
morris
peter
ria

7 rows selected.

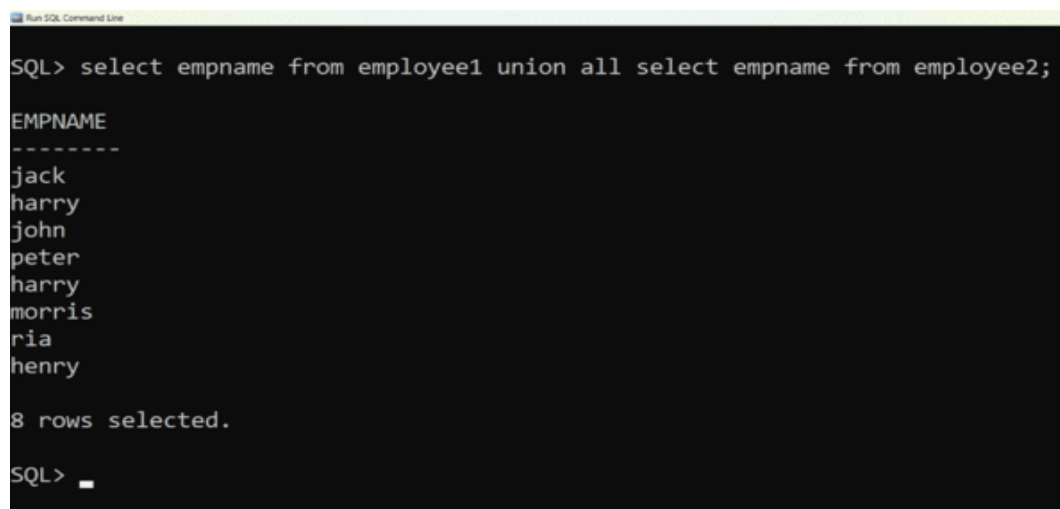
SQL> _
```

UNIONALL

Syntax:

*SQL> select<column_name>from<table_name>unionall select
<column_name>from<table_name>;*

Query:



```
SQL> select empname from employee1 union all select empname from employee2;

EMPNAME
-----
jack
harry
john
peter
harry
morris
ria
henry

8 rows selected.

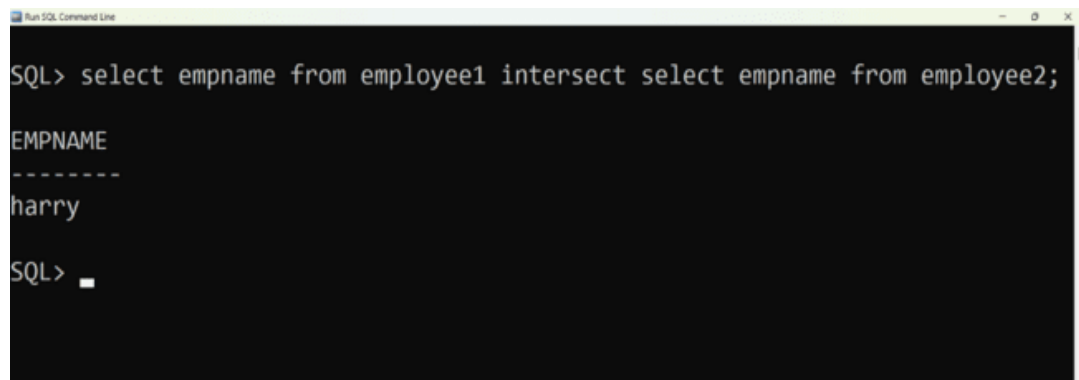
SQL> _
```

INTERSECT

Syntax:

*SQL> select<column_name>from<table_name>intersect select
<column_name>from<table_name>;*

Query:



```
SQL> select empname from employee1 intersect select empname from employee2;

EMPNAME
-----
harry

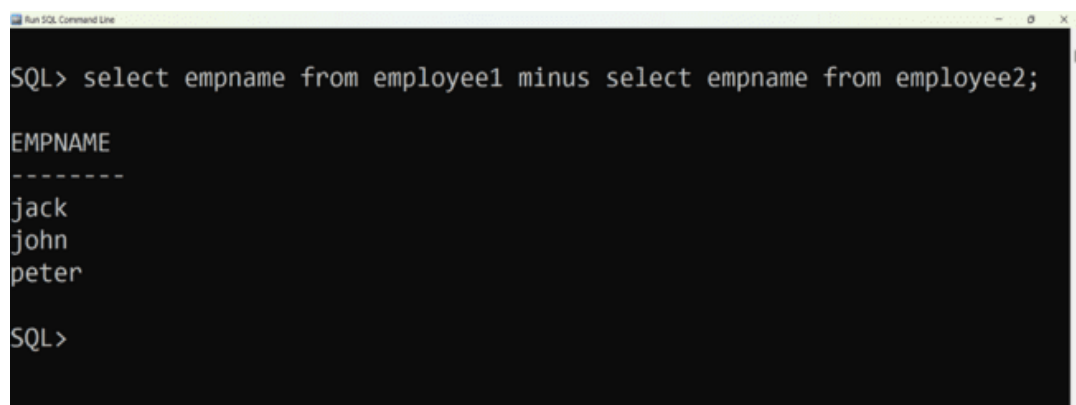
SQL> 
```

MINUS

Syntax:

*SQL> select<column_name>from<table_name>minus select
<column_name>from<table_name>;*

Query:



```
SQL> select empname from employee1 minus select empname from employee2;

EMPNAME
-----
jack
john
peter

SQL> 
```

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SET OPERATIONS

CREATE TABLE employee AND department:

```
mysql> create table Employee(EMP_Name varchar(20), DEPT_NO int);
Query OK, 0 rows affected (0.01 sec)

mysql> insert into Employee(EMP_Name, DEPT_NO) values('Guna', 10);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> create table Department(DEPT_Name varchar(20), DEPT_NO int);
Query OK, 0 rows affected (0.01 sec)

mysql> insert into Department(DEPT_Name, DEPT_NO) values('Manager', 10);
Query OK, 1 row affected (0.00 sec)
```

TABLE STRUCTURE:

Table 1: Employee

```
SQL> select*from Employee;

EMP_NAME      DEPT_NO
-----
guna          10
suman         10
kaushi        20
vetri         30
ash           40
```

Table 2: Department

```
SQL> select*from department;

DEPT_NAME      DEPT_NO
-----
manager        10
salesman       20
staff          30
suman          10
guna           20
SQL> |
```

UNION:

Syntax:

*SQL> select <column_name> from<table_name> union select
<column_name>from<table_name>;*

Query:

```
SQL> select emp_name from employee union select dept_name fr  
om department;  
  
EMP_NAME  
-----  
ash  
guna  
kaushi  
manager  
salesman  
staff  
suman  
vetri  
  
8 rows selected.
```

UNION ALL:

Syntax:

*SQL> select <column_name> from<table_name> union all
select<column_name>from<table_name>;*

Query:

```
SQL> select emp_name from employee union all select dept_name from depar  
tment;  
  
EMP_NAME  
-----  
guna  
suman  
kaushi  
vetri  
ash  
manager  
salesman  
staff  
suman  
guna  
  
10 rows selected.
```

INTERSECTION:

Syntax:

```
SQL> select <column_name> from<table_name> intersect  
select<column_name>from<table_name>;
```

Query:

```
SQL> select emp_name from employee int  
ersect select dept_name from departmen  
t;  
  
EMP_NAME  
-----  
guna  
suman
```

MINUS:

Syntax:

```
SQL> select <column_name> from<table_name> minus  
select<column_name>from<table_name>;
```

Query:

```
SQL> select emp_name from employee minus  
select dept_name from department;  
  
EMP_NAME  
-----  
ash  
kaushi  
vetri  
  
SQL> |
```

CREATE TABLE employee_1:

```
SQL> create table employee_1(EMP_NAME char(25),EMP_ID number(8),EMP_SALARY number(8),EMP_CITY char(15));
Table created.

SQL> insert into employee_1(EMP_NAME,EMP_ID,EMP_SALARY,EMP_CITY)values('mega',34857,40000,'pondy');
1 row created.
```

TABLE STRUCTURE:

Run SQL Command Line

```
SQL> select*from employee_1;
```

EMP_NAME	EMP_ID	EMP_SALARY	EMP_CITY
mega	34857	40000	pondy
ravi	34847	20000	chennai
mani	34237	25000	chennai
venai	34877	40000	madurai
rakul	34127	30000	pondy
raagu	34987	20000	madurai
priya	34097	30000	selam

```
7 rows selected.
```

```
SQL>
```

BETWEEN:

Syntax:

SQL>select <column_name>from<table_name>where<column_name>

BETWEEN value1 AND value2;

Query:

- 1) List all the employee detail with salary in the range 20000 and 30000;

Run SQL Command Line

```
SQL> select emp_name,emp_id from employee_1 where emp_salary between 20000 and 30000;
```

EMP_NAME	EMP_ID
ravi	34847
mani	34237
rakul	34127
raagu	34987
priya	34097

```
SQL>
```


NOT BETWEEN:

Syntax:

SQL>select<column_name>from<table_name>where<field_name>NOT BETWEEN value1 AND value2;

Query:

- 2) List all the employee detail with salary not in the range 20000 and 30000;

 Run SQL Command Line

```
SQL> select emp_name,emp_id from employee_1 where emp_salary not between 20000 and 30000;

EMP_NAME  EMP_ID
-----
mega      34857
venai     34877

SQL> _
```

IN:

Syntax:

SQL>select<column_name>from<table_name>WHERE<column_name>IN (value1,value2,...value n);

Query:

- 3) List the employee detail whose city in pondy and chennai;

 Run SQL Command Line

```
SQL> select emp_name,emp_id from employee_1 where emp_city in ('pondy','chennai');

EMP_NAME  EMP_ID
-----
mega      34857
ravi      34847
mani      34237
rakul     34127

SQL> _
```

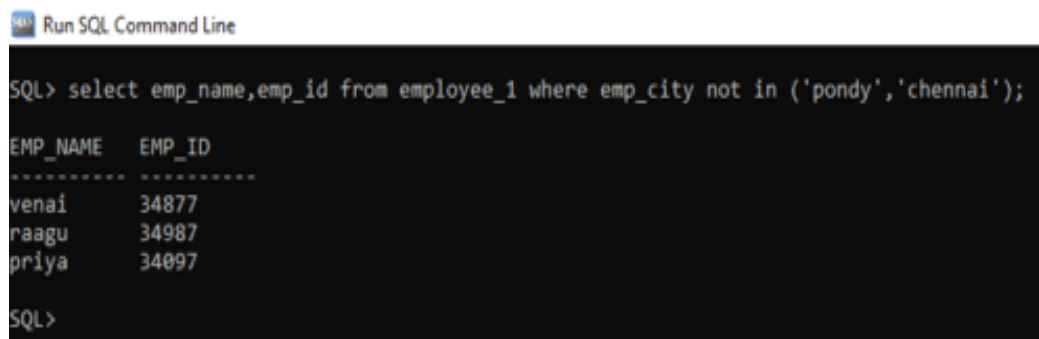
NOT IN:

Syntax:

*SQL>select <column_name>from<table_name>WHERE<field_name> NOT IN
[condition];*

Query:

- 4) List the employee detail whose pndy not in chennai;



The screenshot shows a SQL Command Line window with the following content:

```
SQL> select emp_name,emp_id from employee_1 where emp_city not in ('pondy','chennai');

EMP_NAME  EMP_ID
-----
venai      34877
raagu      34987
priya      34097

SQL>
```

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JOINS

1. Inner Join

- Equi Join
- Non-Equi Join
- Natural Join

2. Outer Join

- Left Outer Join
- Right Outer Join
- Full Outer Join

CREATE TABLE stu AND dept:

```
SQL> create table stu(STU_ID varchar(6),S_NAME char(20),DEPT_ID number(5));  
Table created.  
SQL> insert into stu(STU_ID,S_NAME,DEPT_ID)values(101,'raju',1);  
1 row created.
```

```
SQL> create table dept(DEPT_ID varchar(6),D_NAME char(20));  
Table created.  
SQL> insert into dept(DEPT_ID,D_NAME)values(1,'cse');  
1 row created.
```

TABLE STRUCTURE:

Table 1: stu

```
SQL> select * from stu;  
  
S_ID    S_NAME    DEPT_ID  
-----  
101     raju      1  
102     sai       2  
103     pinky     1  
104     hari      3  
105     jessi     5  
null    null      4  
  
6 rows selected.
```

Table 2: dept

```
SQL> select * from dept;

DEPT_ID  D_NAME
-----
1         cse
2         ece
3         eee
4         mech
5         null
```

INNER JOIN

EQUI JOIN:

Syntax:

SQL>select <column_name> from <table_name> where [condition];

Query:

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu inner join dept on dept.dept_id=stu.dept_id;
```

```
S_ID    S_NAME    DEPT_ID  D_NAME
-----
101     raju      1         cse
102     sai       2         ece
103     pinky     1         cse
104     hari      3         eee
105     jessi     5         null
null    null      4         mech

6 rows selected.
```

NON-EQUI JOIN:

Syntax:

SQL>select <column_name> from <table_name> where [condition];

Query:

```
SQL> select s_name from stu where dept_id<4;
```

```
S_NAME
-----
raju
sai
pinky
hari
```

SELF JOIN:

Syntax:

SQL>select <column_name> from <table_name> where [condition];

Query:

```
SQL> select stu.s_name,dept.dept_id from stu,dept where dept.dept_id=stu.dept_id;
```

```
S_NAME  DEPT_ID
-----  -
pinky    1
raju     1
sai      2
hari     3
null     4
jessi    5

6 rows selected.
```

OUTER JOIN:

LEFT OUTER JOIN:

Syntax:

*SQL>select <column_name> from <table 1> left outer join <table 2>
on [condition];*

Query:

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu left outer join dept on dept.dept_id=stu.dept_id;
```

S_ID	S_NAME	DEPT_ID	D_NAME
103	pinky	1	cse
101	raju	1	cse
102	sai	2	ece
104	hari	3	eee
null	null	4	mech
105	jessi	5	null

6 rows selected.

RIGHT OUTER JOIN:

Syntax:

*SQL>select <column_name> from <table 1> right outer join <table 2>
on [condition];*

Query:

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu right outer join dept on dept.dept_id=stu.dept_id;
```

S_ID	S_NAME	DEPT_ID	D_NAME
101	raju	1	cse
102	sai	2	ece
103	pinky	1	cse
104	hari	3	eee
105	jessi	5	null
null	null	4	mech

6 rows selected.

FULL OUTER JOIN:

Syntax:

*SQL>select <column_name> from <table 1> full outer join <table 2>
on [condition]*

Query:

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu full outer join dept on dept.dept_id=stu.dept_id;
```

S_ID	S_NAME	DEPT_ID	D_NAME
101	raju	1	cse
102	sai	2	ece
103	pinky	1	cse
104	hari	3	eee
105	jessi	5	null
null	null	4	mech

6 rows selected.

CARTESIAN JOIN:

Syntax:

SQL>select table1.column1,table2.column2 from table1,table2;

Query:

```
SQL> select * from stu,dept;
```

S_ID	S_NAME	DEPT_ID	DEPT_ID	D_NAME
101	raju	1	1	cse
101	raju	1	2	ece
101	raju	1	3	eee
101	raju	1	4	mech
101	raju	1	5	null
102	sai	2	1	cse
102	sai	2	2	ece
102	sai	2	3	eee
102	sai	2	4	mech
102	sai	2	5	null
103	pinky	1	1	cse
103	pinky	1	2	ece
103	pinky	1	3	eee
103	pinky	1	4	mech
103	pinky	1	5	null
104	hari	3	1	cse
104	hari	3	2	ece
104	hari	3	3	eee
104	hari	3	4	mech
104	hari	3	5	null
105	jessi	5	1	cse
105	jessi	5	2	ece
105	jessi	5	3	eee
105	jessi	5	4	mech
105	jessi	5	5	null
null	null	4	1	cse
null	null	4	2	ece
null	null	4	3	eee
null	null	4	4	mech
null	null	4	5	null

30 rows selected.

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VIEWS

CREATE TABLE student:

```
SQL> create table student(STUD_ID number(5),STUD_NAME char(25),CGPA float(5),DEPT_ID varchar(5));
Table created.

SQL> insert into student values(100,'Saro',8.1,1);
1 row created.

SQL> insert into student values(101,'Jayaprada',8.5,1);
1 row created.
```

TABLE STRUCTURE:

```
mysql> select * from students;
+-----+-----+-----+-----+
| STUD_ID | STUD_NAME | CGPA | DEPT_ID |
+-----+-----+-----+-----+
| 100 | Saro | 8.1 | 1 |
| 101 | Jayaprakash | 8.5 | 1 |
| 102 | Priya | 8.6 | 1 |
| 103 | Vicky | 7.6 | 1 |
| 104 | Ram | 7.5 | 1 |
| 105 | Mandy | 8.4 | 1 |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

CREATING VIEWS

Syntax:

*SQL>create view <view name> as select column1, column2...column n
from <table name> where [condition];*

Query:

View is created for the table,

```
mysql> CREATE VIEW students_dept1 AS
-> SELECT STUD_ID, STUD_NAME, CGPA, DEPT_ID
-> FROM students
-> WHERE DEPT_ID = 1;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM students_dept1;
+-----+-----+-----+-----+
| STUD_ID | STUD_NAME | CGPA | DEPT_ID |
+-----+-----+-----+-----+
| 100 | Saro | 8.1 | 1 |
| 101 | Jayaprakash | 8.5 | 1 |
| 102 | Priya | 8.6 | 1 |
| 103 | Vicky | 7.6 | 1 |
| 104 | Ram | 7.5 | 1 |
| 105 | Mandy | 8.4 | 1 |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

INSERT:

Syntax:

*SQL>insert into <view_name>(fieldname1,fieldname2....fieldname
n)values(value1,value2....value n);*

Query:

```
mysql> select * from CSE;
+-----+-----+-----+-----+
| stud_id | stud_name | cgpa | dept_id |
+-----+-----+-----+-----+
| 100 | Saro | 8.1 | 1 |
| 101 | JayaPrakash | 8.5 | 1 |
| 102 | Mandy | 7.5 | 1 |
| 103 | Anisha | 9 | 1 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Table after inserting in view,

```
mysql> select * from students;
+-----+-----+-----+-----+
| STUD_ID | STUD_NAME | CGPA | DEPT_ID |
+-----+-----+-----+-----+
| 100 | Saro | 8.1 | 1 |
| 101 | Jayaprakash | 8.5 | 1 |
| 102 | Priya | 8.6 | 1 |
| 103 | Vicky | 7.6 | 1 |
| 104 | Ram | 7.5 | 1 |
| 105 | Mandy | 8.4 | 1 |
| 106 | Anmol | 9 | 2 |
| 107 | Akash | 7.5 | 2 |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

UPDATING A VIEW:

Syntax:

SQL>update <view_name> where [condition];

Query:

```
mysql> update CSE set cgpa=9 where stud_id = 100;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from CSE;
+-----+-----+-----+-----+
| stud_id | stud_name | cgpa | dept_id |
+-----+-----+-----+-----+
| 100 | Saro | 9 | 1 |
| 101 | JayaPrakash | 8.5 | 1 |
| 102 | Mandy | 7.5 | 1 |
| 103 | Anisha | 9 | 1 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Table after updating the view,

```
mysql> select * from students;
+-----+-----+-----+-----+
| STUD_ID | STUD_NAME | CGPA | DEPT_ID |
+-----+-----+-----+-----+
| 100 | Saro | 9 | 1 |
| 101 | Jayaprakash | 8.5 | 1 |
| 102 | Mandy | 7.5 | 1 |
| 103 | Anisha | 9 | 1 |
| 104 | Ram | 7.5 | 2 |
| 105 | Mandy | 8.4 | 2 |
| 106 | Anmol | 9 | 2 |
| 107 | Akash | 7.5 | 2 |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

DELETE:

Syntax:

SQL> delete from <view_name> where [condition];

Query:

```
mysql> delete from CSE where stud_id = 103;
Query OK, 1 row affected (0.00 sec)

mysql> select * from CSE;
+-----+-----+-----+-----+
| stud_id | stud_name | cgpa | dept_id |
+-----+-----+-----+-----+
| 100 | Saro | 9 | 1 |
| 101 | JayaPrakash | 8.5 | 1 |
| 102 | Mandy | 7.5 | 1 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Table after deleting in view,

```
mysql> select * from students;
+-----+-----+-----+-----+
| STUD_ID | STUD_NAME | CGPA | DEPT_ID |
+-----+-----+-----+-----+
| 100 | Saro | 9 | 1 |
| 101 | Jayaprakash | 8.5 | 1 |
| 102 | Mandy | 7.5 | 1 |
| 104 | Ram | 7.5 | 2 |
| 105 | Mandy | 8.4 | 2 |
| 106 | Anmol | 9 | 2 |
| 107 | Akash | 7.5 | 2 |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

DROP:

Syntax:

SQL>drop view<view_name>;

Query:

 SQL Plus

```
SQL> drop view CSE;
```

```
View dropped.
```

After the view is dropped the table will be remind the same,

```
mysql> select * from CSE;
ERROR 1146 (42S02): Table 'jazxd.cse' doesn't exist
mysql> select * from students;
+-----+-----+-----+-----+
| STUD_ID | STUD_NAME | CGPA | DEPT_ID |
+-----+-----+-----+-----+
| 100 | Saro | 9 | 1 |
| 101 | Jayaprakash | 8.5 | 1 |
| 102 | Mandy | 7.5 | 1 |
| 104 | Ram | 7.5 | 2 |
| 105 | Mandy | 8.4 | 2 |
| 106 | Anmol | 9 | 2 |
| 107 | Akash | 7.5 | 2 |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

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SUB QUERIES

CREATE TABLE student AND database:

```
mysql> create table data(STU_NAME varchar(10), STU_ROLL int, STU_LOCATE varchar(10));
Query OK, 0 rows affected (0.01 sec)

mysql> insert into data(STU_NAME, STU_ROLL, STU_LOCATE) values('Ravi', 1123, 'Chennai');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> create table STUdata(STU_NAME varchar(10), STU_ROLL int, STU_SEC varchar(10));
Query OK, 0 rows affected (0.01 sec)

mysql> insert into STUdata(STU_NAME, STU_ROLL, STU_SEC) values('Ravi', 1123, 'A');
Query OK, 1 row affected (0.00 sec)
```

TABLE STRUCTURE:

 Run SQL Command Line

```
SQL> select*from database;

STU_NAME  STU_ROLL  STU_LOCATI
-----
RAVI      1123      chennai
RAJ       1124      madurai
RAM       1125      salem
SAM       1126      theni
MANI      1127      chennai

SQL> select*from student;

STU_NAME  STU_ROLL  STU_SEC
-----
RAVI      1123      A
RAJ       1124      B
RAM       1125      A

SQL>
```

SELECT:

Syntax:

SQL> SELECT column_name

FROM table_name

WHERE column_name expression operator

(SELECT column_name from table_name WHERE ...);

Query:

 Run SQL Command Line

```
SQL> select stu_name,stu_location from database where stu_roll in (select stu_roll from student where stu_sec='A');  
  
STU_NAME  STU_LOCATI  
-----  
RAVI      chennai  
RAM       salem  
  
SQL>
```

INSERT

Syntax:

SQL> INSERT INTO table_name (column1, column2, column3,...)

*SELECT **

FROM table_name

WHERE VALUE OPERATOR

Query:

```
Run SQL Command Line

SQL> select*from student1;

NAME          ROLL_NO    LOCATION    PHONENUMBER
-----
RAVI          1212       CHENNAI     9003733628
MOTHI         1232       THENI       7397080037
MADHAN        1242       MADURAI     9790585064
MORISH        1282       THENI       1234080037

SQL> select*from student2;

NAME          ROLL_NO    LOCATION    PHONENUMBER
-----
RAHUL         4212       KERALA      9093433628
MASHA         1242       MADURAI     9792285064
MONISHA       1442       CHENNAI     9792285234

SQL> insert into student1 select*from student2;

3 rows created.

SQL> select *from student1;

NAME          ROLL_NO    LOCATION    PHONENUMBER
-----
RAVI          1212       CHENNAI     9003733628
MOTHI         1232       THENI       7397080037
MADHAN        1242       MADURAI     9790585064
MORISH        1282       THENI       1234080037
RAHUL         4212       KERALA      9093433628
MASHA         1242       MADURAI     9792285064
MONISHA       1442       CHENNAI     9792285234

7 rows selected.

SQL>
```


DELETE:

Syntax:

SQL>DELETE FROM TABLE_NAME

WHERE VALUE OPERATOR

(SELECT COLUMN_NAME

FROM TABLE_NAME

WHERE condition);

```
SQL> delete from student2 where roll_no in (select roll_no from student1 where location='CHENNAI');
1 row deleted.
SQL> select*from student2;
```

NAME	ROLL_NO	LOCATION	PHONENUMBER
MASHA	1242	MADURAI	9792285064
RAKUL	1542	KERALA	1234564552

```
SQL>
```

UPDATE

Syntax:

SQL> UPDATE table

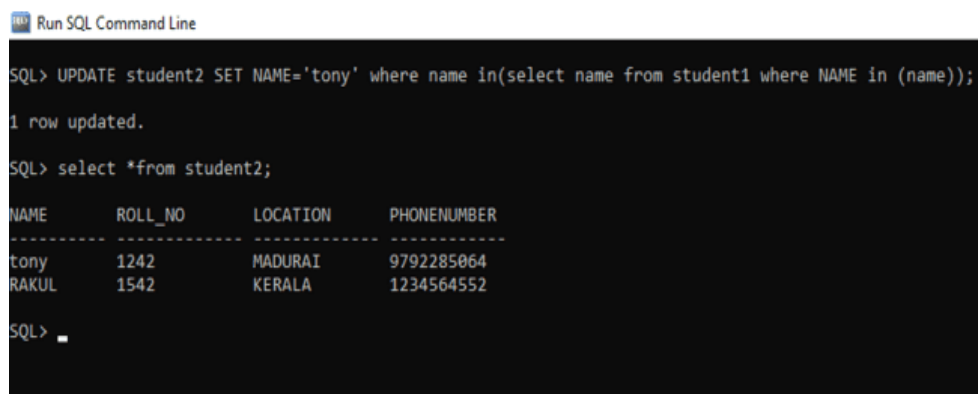
SET column_name = new_value

WHERE VALUE OPERATOR

(SELECT COLUMN_NAME

FROM TABLE_NAME

WHERE condition);



Run SQL Command Line

```
SQL> UPDATE student2 SET NAME='tony' where name in(select name from student1 where NAME in (name));
1 row updated.
SQL> select *from student2;
```

NAME	ROLL_NO	LOCATION	PHONENUMBER
tony	1242	MADURAI	9792285064
RAKUL	1542	KERALA	1234564552

SQL> _

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ADDITION OF TWO NUMBERS

PROGRAM:

SQL>set serveroutput on;

```
declare
x integer;
y integer;
z integer;
begin
x:=&x;
y:=&y;
z:=x+y;
dbms_output.put_line('The sum of two numbers is'||z);
end;
/
```

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FACTORIAL OF A NUMBER

PROGRAM:

SQL> set serveroutput on;

```
declare
f number:=1;
i number;
n number:=&n;
begin
for i in 1..n
loop
f:=f*i;
end loop;
dbms_output.put_line('the factorial of given number is:'||f);
end;
/
```

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SUM OF SERIES

PROGRAM:

SQL> set serveroutput on;

```
declare
n number:=&n;
i number;
begin
i:=n*(n+1);
n:=i/2;
dbms_output.put_line('the sum of series of given number is:'||n);
end;
/
```

REG NO: 22TD0006 NAME: ARUL SHRINIVAS A YEAR/SEM/SEC: Y3/S5/B

FIBONACCI SERIES

PROGRAM:

SQL>set serveroutput on;

```
declare
i number;
c number;
n number:=&n;
a number:=-1;
b number:=1;
begin
for i in 1..n
loop
c:=a+b;
dbms_output.put_line(c);
a:=b;
b:=c;
end loop;
end;
/
```

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PALINDROME SERIES

PROGRAM:

SQL>set serveroutput on;

```
declare
len number;
a integer;
str1 varchar(10):='&str1';
str2 varchar(10);
begin
len:=length(str1);
a:=len;
for i in 1..a
loop
str2:=str2||substr(str1,len,1);
len:=len-1;
end loop;
if(str1=str2)then
dbms_output.put_line(str1||' is a palindrome');
else
dbms_output.put_line(str1||' is not a palindrome');
end if;
end;
/
```

REG NO: 22TD0006 NAME: ARUL SHRINIVAS A YEAR/SEM/SEC: Y3/S5/B

PACKAGES

SOURCE CODE:

PACKAGE EXCEPTION:

```
create or replace package packs is
procedure display;
function prime(n number) return varchar;
function odd(n number) return varchar;
function positive(n number) return varchar;
end packs;
```

PACKAGE BODY:

```
create or replace package body packs as
procedure display is
avarchar(15);
m number;
begin
m:=&m;
a:=prime(m);
if a=prime(m)then
dbms_output.put_line(m||'is a prime');
else
dbms_output.put_line(m||'is not prime');
end if;
```



```

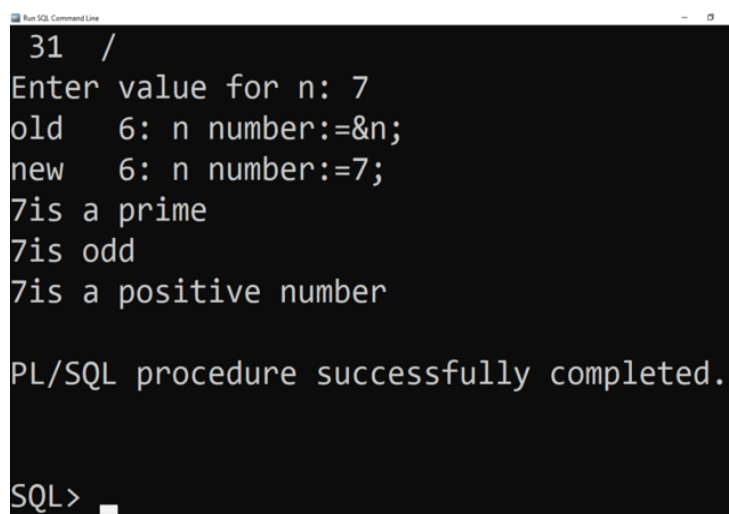
a:=odd(m);
if a='odd' then
dbms_output.put_line(m||'is odd');
else
dbms_output.put_line(m||'is even');
end if;
a:=positive(m);
if a='positive' then
dbms_output.put_line(m||'is a positive number');
else
dbms_output.put_line(m||'is a negative number');
end if;
end display;
function prime(n number) return varchar is
i number;
flag number;
begin
flag:=0;
for i in 2..n/2
loop
if mod(n,i)=0 then
flag:=1
return 'not prime';
end if;
end loop;
if flag=0 then
return 'prime';
end if;

```

```
end prime;

function odd(n number) return varchar is
begin
if n>0 then
return 'positive';
else
return 'negative';
end if;
end positive;
end packs;
```

OUTPUT:



```
Run SQL Command Line
31 /
Enter value for n: 7
old 6: n number:=&n;
new 6: n number:=7;
7is a prime
7is odd
7is a positive number

PL/SQL procedure successfully completed.

SQL> █
```

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CURSOR

SOURCE CODE:

DELETING THE RECORD USING CURSOR

```
SQL> declare
    cursor c is select * from vechicle1;
    a c % rowtype;
    n number:=0;
begin
    open c;
loop
    fetch c into a;
    exit when c% notfound;
    if a.vehicle_no=521 then
        delete from vehicle1 where vehicle_no=521;
        n:=n+1;
    end if;
end loop;
dbms_output.put_line('deleted record'||n);
dbms_output.put_line('remaining records'||(c % rowcount-n));
close c;
commit;
end;
/
```

OUTPUT:

```
VEHICLE_NO VEHIC SOURCE      DESTINATIO SEAT_COST  ARR_TIME
-----
521 duke   pondy      cuddalore  52         12
623 prtc   pondy      banglore   35         2
632 mnc    bangalore  chennai    23         1

SQL>
```

```
PL/SQL procedure successfully completed.
```

```
SQL> select * from vechicle1;
```

```
VEHICLE_NO VEHICLE SOURCE      DESTINATIO SEAT_COST  ARR_TIME
-----
623 prtc   pondy      banglore   35         2
632 mnc    bangalore  chennai    23         1

SQL> _
```